

CUMBERLAND, WI, USA

WMO: 720929

Lat: 45.506N

Lon: 91.981W

Elev: 378

StdP: 96.86

Time Zone: -6.00 (NAC)

Period: 11-19

WBAN: 00316

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.2	(c) -24.8	(d) -31.3	(e) 0.2	(f) -26.1	(g) -28.7	(h) 0.3	(i) -23.4	(j) 10.9	(k) -8.2	(l) 9.8	(m) -7.8	(n) 2.2	(o) 310	(p) 0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	28.9	21.5	27.4	20.2	26.2	19.2	23.0	27.2	21.9	25.9	20.7	24.0	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.5	26.2	20.3	15.7	24.3	19.7	15.1	23.6	70.1	27.2	65.8	26.5	61.4	24.9	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.5	7.6	DB	-31.0	31.0	2.9	1.7	-33.1	32.2	-34.8	33.2	-36.4	34.2	-38.5	35.4
			WB	-31.2	24.2	2.8	1.4	-33.3	25.2	-34.9	26.1	-36.5	26.9	-38.6	27.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-10.5	-9.6	-1.8	4.5	12.4	17.7	19.9	18.2	14.6	7.1	-0.8	-7.1
	DBStd	12.04	7.72	7.02	7.13	5.21	4.68	3.19	2.93	2.84	4.48	4.66	5.69	6.75
	HDD10.0	2746	636	549	372	177	28	0	0	0	11	117	325	532
	HDD18.3	4851	894	782	625	416	190	48	17	39	127	349	574	790
	CDD10.0	1088	0	0	6	12	103	231	308	253	149	26	1	0
	CDD18.3	152	0	0	0	0	7	28	66	33	16	1	0	0
	CDH23.3	1251	0	0	1	1	93	234	547	248	117	10	0	0
	CDH26.7	226	0	0	0	0	19	45	109	34	18	1	0	0
Wind	WSAvg	3.3	3.7	3.6	3.6	3.9	3.6	3.2	2.6	2.2	2.8	3.3	3.5	3.3
Precipitation	PrecAvg	822	25	23	42	68	91	107	107	98	94	85	46	29
	PrecMax	1111	58	55	82	189	187	152	184	215	200	175	147	59
	PrecMin	574	5	3	18	13	33	33	22	30	24	28	5	7
	PrecStd	124	14	14	19	38	39	35	42	43	47	42	37	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	7.8	20.2	21.4	29.7	30.2	32.2	30.0	28.9	25.0	16.1	7.0
		MCWB	2.0	5.7	15.6	13.4	18.0	21.7	25.1	22.1	20.6	14.7	10.4	5.2
	2%	DB	2.1	3.7	15.1	18.7	26.1	27.6	29.2	27.4	27.0	21.0	12.4	3.8
		MCWB	0.8	1.2	11.7	11.0	16.7	19.5	22.6	20.6	20.0	13.4	8.9	2.3
	5%	DB	0.8	1.3	11.2	16.1	22.8	26.1	27.5	26.1	24.8	17.4	9.9	2.2
		MCWB	0.1	0.0	7.8	9.0	15.4	18.6	20.8	19.4	18.1	12.5	6.9	1.4
	10%	DB	-0.1	0.1	7.3	12.9	21.0	24.7	26.3	24.9	22.3	14.9	7.2	0.3
		MCWB	-0.6	-0.8	4.6	7.3	14.1	17.5	19.6	19.0	17.6	10.5	4.9	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.7	15.9	14.0	20.7	23.5	26.2	23.5	22.7	16.9	12.1	6.0
		MCDB	4.0	7.9	20.1	20.2	26.4	28.5	30.6	28.0	26.4	21.2	14.1	6.4
	2%	WB	0.7	1.8	12.9	11.6	18.1	21.6	23.5	22.2	21.0	14.6	9.1	2.7
		MCDB	1.4	3.2	15.0	16.4	22.3	25.9	27.6	26.2	23.9	19.3	12.5	3.5
	5%	WB	0.0	0.2	7.6	10.0	17.0	19.8	22.2	20.9	19.6	12.8	6.8	1.0
		MCDB	0.4	0.9	9.9	14.7	20.9	24.6	26.4	23.4	23.2	16.7	9.1	1.7
	10%	WB	-1.0	-0.9	4.4	8.1	15.6	18.7	21.1	19.8	18.4	11.4	4.9	0.1
		MCDB	-0.5	-0.1	7.1	12.6	19.5	22.8	24.6	23.8	21.5	14.0	7.2	0.4
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.7	9.3	9.6	11.4	10.4	10.8	10.9	11.2	9.3	7.6	7.0
		MCDBR	7.5	8.8	12.4	15.0	14.5	12.4	11.7	12.2	13.2	14.0	12.1	7.4
		MCWBR	6.4	7.4	7.9	8.6	7.2	6.1	6.1	6.4	7.0	8.0	8.0	6.4
	5% WB	MCDBR	7.4	7.8	11.3	13.4	11.8	10.5	10.6	10.4	10.5	12.0	10.9	6.0
		MCWBR	6.7	6.8	7.9	8.4	6.9	6.1	6.3	6.1	6.3	7.6	7.8	5.2
Clear-Sky Solar Irradiance	taub	0.252	0.270	0.310	0.352	0.384	0.381	0.372	0.384	0.358	0.313	0.283	0.260	
	taud	2.390	2.336	2.356	2.358	2.307	2.346	2.393	2.374	2.424	2.537	2.515	2.425	
	Ebn at Noon	876	920	923	907	884	885	889	865	860	863	833	826	
	Edh at Noon	71	92	105	115	126	122	115	113	98	76	64	62	
All-Sky Solar Radiation	RadAvg	1.66	2.77	3.89	4.60	5.32	5.88	6.17	5.15	4.00	2.51	1.55	1.15	
	RadStd	0.18	0.24	0.35	0.45	0.42	0.45	0.23	0.37	0.26	0.31	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page